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Representing and Learning Stress: A MaxEnt Framework for Comparing Learning Across Grammatical Theories

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Draft of Oct. 30, 2024

Keywords: Stress, prosody, learning, typology, maximum entropy

Abstract

This paper examines learning of word stress patterns across grammatical theories, within a Maximum Entropy learning framework. This framework achieves results comparable to the state-of-the-art on a well-studied test of stress learning, which uses a set of foot-based constraints as the grammatical theory. The typological coverage of this constraint set has not previously been examined. A computational study shows that it can represent about half of a sample of attested languages, and that an alternative foot-based constraint set, as well as a grid-based one, succeed in capturing more of them. In terms of learning, the foot-based constraint sets have high rates of success on the languages they can represent, even in face of the structural ambiguity introduced by feet. Comparisons of learning speed on different language types across grammatical theories reveal differences amongst them that would not be evident in a comparison of their representational capacity alone.

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1. Introduction and summary of results

A fundamental goal of generative linguistics is to provide a theory of grammar that is not only sufficiently expressive to capture attested linguistic phenomena, but is also able to help explain language acquisition by structuring the hypothesis space in a way that facilitates learning (Chomsky, 1959, 1980). In this paper, we propose and explore a way of making explicit learnability comparisons across theories of grammar. We take a fixed theory of learning, and use it to test the success and efficiency of learning when different theories of grammar are used in combination with the learning theory.

Maximum Entropy Grammar (MaxEnt) is a variant of Prince and Smolensky’s (1993/2004) Optimality Theory (OT) that uses weighted constraints to define a probability distribution over the candidates in a tableau. In introducing it, Goldwater and Johnson (2003) point out that “[t]he mathematical properties of the model have been well-studied, it has been shown to be useful for learning in a variety of domains, and in general there is a wide literature available”. As such, it is a useful general framework for comparing learning under varying assumptions about the nature of the grammar. In this paper, we compare learning under two approaches to the representation of metrical structure (feet vs. grids), and also explore some of the consequences of varying the constraint sets within each one. We are making software created for this research freely accessible in hopes that it will facilitate further exploration of this framework (Lee and Pater, 2024; Prickett and Pater, 2024).

We start with the Tesar and Smolensky (2000, henceforth TS) foot-based constraint set, and their test set of 124 stress patterns, which are proposed as an initial exploration of what TS call the hidden structure problem, also known as the problem of structural ambiguity (see Jarosz 2019 for an overview of work on this topic). In a foot-based theory, a single stress pattern can correspond to multiple prosodifications (e.g. [batáma] parsed with a right-headed iamb [(batá)ma] or left-headed trochee [ba(táma)]). In the TS learning problems, the learner is told only which stress pattern is correct in a given language, and must infer the prosodification. Learning of this test set with these constraints has been well-studied, and we introduce MaxEnt learning of stress with hidden structure by providing results that are comparable to prior work. We find that the success rate is comparable to the state-of-the-art results presented in Jarosz (2013) and Jarosz (2015).

Many of the patterns that our learner has difficulty with appear to be unattested. Most of them have a shared characterization with this constraint set as iambic languages that use non-finality to keep stress off the final syllable. The absence of these patterns is a mystery in a foot-based theory, and this result suggests that it may have a learning-based explanation.

We then turn to the question of how well the TS constraint set captures the attested typology of stress systems, and explore learning with it and with alternative constraint

sets that capture patterns that escape the TS constraints. The question of what exactly the “attested typology” consists of is a large and contentious one, so we limit ourselves here to an examination of 61 patterns provided in StressTyp2 (Goedemans et al., 2015) as finite state transducers. We use these FSTs to create strings of stressed and unstressed syllables, mimicking the dataset format in the TS test set, and then we assign constraint violations with each of the constraint sets we examine.

The TS constraint set is based on the proposals in Prince and Smolensky (1993/2004) and McCarthy and Prince (1993). The TS formulation of the constraints that place main stress, and the constraint that prefers left-headed (trochaic) feet both also appear in alternative versions in that earlier work. We find that adopting these alternatives allows us to capture a greater portion of the attested typology, as does adding a nonfinality constraint that targets final main stress. The TS constraints can represent 31 of the 61 StressTyp languages, while the revised TS set adds 12 more, for a total of 43.

We examine the differences between the constraint formulations in some detail with case studies of Latin and Québécois, both of which are captured only with the revisions to the TS constraint set. In assessing the representational capacity of the constraint sets, we make heavy use of a new implementation of the Linear Programming method of Potts et al. (2010), which adds the ability to operate with hidden structure (Lee and Pater, 2024).

An important general point that emerges from our study of the learning of Latin and Québécois with differing foot-based constraint sets is that a smaller hypothesis space does not always yield improved learnability. We show that a larger constraint set, which yields a wider set of representational possibilities for Latin, also yields better learning results. It is also not that case that a larger hypothesis space always yields better learning. Rather, it is the structure of the space that matters, whether it tends to lead learners into local optima.

In addition to comparisons amongst foot-based theories, we examine learning with constraint sets that make fundamentally different assumptions about the representation of metrical structure. Gordon (2002) proposes a set of grid-based constraints that captures a typology of quantity insensitive stress. We examine two grid-based constraint sets. The first uses the constraints from Gordon (2002), supplemented with a Weight-to-Stress constraint that allows it to capture some of the quantity sensitive stress patterns in StressTyp2. The second changes the definition of the constraints that place main stress to be parallel to the ones used in the revised version of the TS constraint set.

In terms of the typological coverage on the 61 stress patterns from StressTyp, we find that the Gordon (2002) constraints plus Weight-to-Stress capture all 31 languages that the TS constraint set does, plus an additional 10, for a total of 41. Five of the additional languages are ones that our revised TS set can represent, with 5 others that are only represented by the Gordon constraints. The revised TS set captures 7 languages that escape the Gordon constraints. These typological results highlight the fact that further research is needed on the representational capacity of OT constraint sets.

Gordon (2011, p. 163) points to the need to infer hidden structure in learning a foot-based analysis as a potential advantage of a grid-based approach. Metrical grids are in a one-to-one relationship with strings of syllables with their stress levels. Since there is no structural ambiguity, the learning algorithms we use are guaranteed to converge on

a correct grammar (see section 3.1 for our definition of “correctness” in a probabilistic setting).

There are (at least) two questions to ask in comparing learning across feet and grids. The first is whether a learner operating with a foot-based grammar will fail to find a correct grammar. To address this question, we used the same learning algorithm and constraint set as in our study of the TS test set, and found similarly high rates of success.

The second question is whether learning with feet or grids is in general more efficient, and also what patterns each theory predicts are easier to learn. We examine this question in terms of the number of epochs it takes to reach a correct grammar. We find there is no clear winner between feet and grids in terms of speed of learning, and that the details of the constraint sets can dramatically affect the results.

We tested two grid-based and foot-based constraint sets, each with relatively minor differences between them. In each case, which of the constraint sets results in more efficient learning depends on whether we examine the quantity sensitive (QS) or quantity insensitive (QI) languages in our test set. In the case of the foot-based constraints, the original TS constraint set does better on the QI languages, while our revised constraint set does better on the QS languages. Since all of the languages in the test set we used for this comparison can be represented with either constraint set, this provides a concrete example where the study of learning can provide added insight into the adequacy of grammatical formulations.

2. Background

2.1. Hidden structure learning with MaxEnt

In the TS learning problems, prosodic structure is *hidden* in the sense that the target learning data consist only of strings of heavy and light syllables with stress indicated, e.g. a string of three light syllables with medial stress, represented in the TS notation as [L L1 L]. A stress pattern for a given type of word will usually be consistent with multiple prosodifications. The learner is supplied with fully prosodified candidates and their constraint violations, and an indication of which candidates are consistent with each learning datum (e.g. that both the iambic [(L L1)L] and the trochaic [L(L1 L)] are consistent with the [L L1 L] learning datum). The goal of learning is to find a ranking of the constraints that will make one of the consistent candidates optimal for every learning datum.

When a learner is supplied with an unprosodified string as a piece of learning data, it must assign it a full structure in order to update the grammar, since the constraint violations are at least partially unknown. In the TS proposal, the learner parses the learning datum using its current grammar, providing a full structure that is the basis for updating the constraint ranking (*Robust Interpretive Parsing*; see Jarosz 2013, 2015 for alternatives). The MaxEnt approach to hidden structure learning that we adopt (see Pater et al. 2012) uses the current probabilities of the consistent candidates to estimate a violation vector for the update of the constraint weights.

The assignment of candidate probabilities by a MaxEnt grammar, and the creation of the violation vector for a learning datum are illustrated in Example (1). The constraint names are from TS; definitions appear in §3 below. AllFeet-R and Nonfin conflict

on whether the final syllable should be footed, and Iambic and FootNonfin (the TS Trochee constraint) conflict on whether the foot should be right- or left-headed. Constraint violations are shown as negative integers, and the current constraint weights are shown underneath each constraint. The Harmony (H) is the weighted sum of constraint violations, as in Harmonic Grammar (Smolensky and Legendre, 2006). In MaxEnt, the probability (p) of a candidate is proportional to the exponential of its Harmony; the probability of each of the four candidates is shown in the rightmost column. These four candidates are all of the possible prosodifications of the trisyllabic string, if we assume strictly binary feet. The last two candidates are the two prosodifications corresponding to stress on the medial syllable. The second set of probabilities for those candidates provided in parentheses are normalized across just them, rather than the whole candidate set; we use these in constructing a violation vector for the ambiguously prosodified learning datum L L1 L.

- (1) Candidates and their probabilities for [L L L], and the violation vector for learning datum [L L1 L] based on probabilities of consistent prosodifications

	AllFeet-R 3	Nonfin 1	Iambic 1	FootNonfin 1	H	p
a. L(L L1)		-1		-1	-2	0.44
b. (L1 L)L	-1		-1		-4	0.06
c. L(L1 L)		-1	-1		-2	0.44 (0.88)
d. (L L1)L	-1			-1	-4	0.06 (0.12)
L L1 L	-0.12	-0.88	-0.88	-0.12		

The assigned violation vector for [L L1 L] is shown in the bottom row. It is the probability-weighted sum of violations of the violation vectors of the consistent candidates (1c.) and (1d.), using the parenthesized probabilities. The lesser penalty assigned by Nonfin and Iambic reflects the greater probability of candidate (1c.), which violates those constraints. Since the violation vector is the representation, we can say that the representation assigned to the learning datum is a blend of the representations of the consistent prosodifications, with the strength of each reflecting its probability.

The learning data are a probability distribution over the unprosodified strings (termed the overt forms in TS). The learning objective is to maximize the likelihood of the learning data, or equivalently, to minimize the difference between the probabilities assigned by the grammar and the probabilities in the learning data (mathematically equivalent when the difference is calculated as K-L divergence). To calculate the probability of a learning datum, we sum the probabilities of the consistent candidates. With the constraint weights in (1), the learning datum [L L1 L] has probability 0.50. Its probability will be maximized as that sum approaches 1, which would happen for example as we raised the weights of AllFeet-R and FootNonfin relative to the other constraints. This would make [L(L1 L)] more and more probable since it violates neither constraint, while the other candidates all violate one or both of them.

To show how a learner can use this inferred structure to update the constraint weights and converge on a correct grammar, we will use Staubs and Pater's (2013) formulation of the widely used Gradient Descent algorithm, which highlights its relationship to other learning models that have been used in the phonological literature, such as the

Gradual Learning Algorithm for OT (Boersma (1998)) and HG (Boersma and Pater (2016)).

If we assume that the learning data give [L L1 L] probability 1, the observed data is the violation vector in the last row of (1), repeated in the Observed row of (2). To create a vector for the update of the constraint weights we subtract a vector representing the learner's expectation from the vector representing the observed data. In OT terms, we subtract the Loser's violations from the Winner's, hence the W and L in the table. The learner's expectation is the probability weighted sum over the whole candidate set, using the probabilities generated by the grammar, which results in the values shown in the Expected row of (2). These values can be checked by taking the probability weighted sum of the four candidate vectors in (1).

- (2) Calculation of update vector as Observed – Expected (Winner – Loser)

	AllFeet-R	Nonfin	Iambic	FootNonfin
Observed (W)	-0.12	-0.88	-0.88	-0.12
Expected (L)	-0.12	-0.88	-0.50	-0.50
Obs. - Exp.	0	0	-0.38	+0.38

The update vector is added to the current constraint weights, scaled by a learning rate. This will increase the weight of FootNonfin and decrease the weight of Iambic, thus increasing the probability of the trochaic candidates. If we assume the learning rate is 1, we get the updated weights and new probabilities in (3).

- (3) Updated weights, harmonies and probabilities

	AllFeet-R 3	Nonfin 1	Iambic 0.62	FootNonfin 1.38	H	<i>p</i>
a. L(L L1)		-1		-1	-2.38	0.28
b. (L1 L)L	-1		-1		-3.62	0.08
c. L(L1 L)		-1	-1		-1.62	0.60 (0.94)
d. (L L1)L	-1			-1	-4.38	0.04 (0.06)

The Observed and Expected vectors based on these updated probabilities are shown in (4), with Observed again being the weighted sum over the last two candidates using the parenthesized probabilities, and Expected again the probability weighted sum over the whole tableau. The resulting second update vector increases the weight of AllFeet-R relative to Nonfin, as well as Iambic relative to FootNonfin.

- (4) Calculation of second update vector

	AllFeet-R	Nonfin	Iambic	FootNonfin
Observed (W)	-0.06	-0.94	-0.94	-0.06
Expected (L)	-0.12	-0.88	-0.68	-0.32
Obs. - Exp.	0.06	-0.06	-0.26	0.26

With this and subsequent updates the learner will assign more and more probability to [L(L1 L)], getting arbitrarily close to 1 with an increasing number of updates ($p = 1$ is impossible since there will always be some probability assigned to the other candidates).

It is important to note that even though it is successful here, this learner, like other learners capable of operating with hidden structure, is not guaranteed to converge on a correct grammar. To show this, we will now provide a simple example of a local optimum that traps this learner, building on Boersma and Pater's (2016) example, which uses related grammatical assumptions and learning algorithms (an explanation of the relationship between our approach and Boersma and Pater's 2016 appears just after the presentation of the local optimum). We have added the two candidates for a bisyllabic word, which exhaust the space of possibilities with strictly binary feet. The learning data are [L L1 L] and [L L1], both with probability 1 for that stressing of that word shape. With these weights, [L(L1 L)] has probability approaching 1 as the full representation for medial stress, and the violation vector for [L L1 L] will resemble to some level of precision the vector in (5c.). The problem is that this trochaic parsing is inconsistent with the iambic foot needed to correctly parse [L L1].

(5) Local optimum for L L1 L and L L1 as target data

	AllFeet-R 20	Nonfin 1	Iambic 1	FootNonfin 1	H	<i>p</i>
a. L(L L1)		-1		-1	-2	0.50
b. (L1 L)L	-1		-1		-21	0
c. L(L1 L)		-1	-1		-2	0.50 (1)
d. (L L1)L	-1			-1	-21	0
e. (L L1)		-1		-1		0.50 (1)
f. (L1 L)		-1	-1			0.50

To do an update over multiple tableaux, we can take a probability weighted sum over them in calculating both the observed and the expected vectors. If we give the two learning data [L L1 L] and [L L1] equal probability, the observed and expected vectors are as in (6). Note that these give equal representation to the iambic and trochaic parses (the constraints that each violate have an equal score of -0.5), and both give final footing the same representation (a score of -1 on the constraint it violates, and 0 on the constraint violated by initial footing on the trisyllable). This table also shows that the update will not change the weights at all, at least at the degree of precision on the weights we are using here (i.e. rounding to two decimal points). This is a local optimum in that this set of weights produces the best result in the space of solutions that the learner can access given the algorithm it is using. In essence, the learner's belief that the foot lies at the right edge is so strong that it cannot consider the possibility that it is not.

(6) Observed/expected vectors and resulting weight updates at a local optimum

	AllFeet-R	Nonfin	Iambic	FootNonfin
Observed (<i>W</i>)	0	-1	-0.5	-0.5
Expected (<i>L</i>)	0	-1	-0.5	-0.5
Obs - Exp.	0	0	0	0

As mentioned above, this learning algorithm is very similar to ones used in the prior phonological literature. If it were run on-line, where each learning update is for a single tableau sampled from the learning data, it would be very similar to

using a MaxEnt grammar in Boersma and Pater’s (2016) hidden structure learning set-up, which combines TS’s Robust Interpretive Parsing with the HG-GLA, whose update rule is equivalent to the one above. The only difference would be that we use probability weighted summation over consistent candidates rather than sampling from the distribution over them. A batch approach also eliminates on-line sampling from the set of tableaux; each learning update is over a probability weighted sum of the whole dataset rather than over a single tableau (see further Pater and Staubs, 2013). In the fully batch learning that we use here, which eliminates these two kinds of sampling, the outcome for a given set of initial weights is deterministic. This is convenient for learning experiments since it eliminates the need to average over multiple runs.

Given the objective function (maximizing likelihood or minimizing error), there are various possible choices for an algorithm to find weights that optimize it – Gradient Descent is just one. Gradient Descent is the algorithm of choice in our comparison of the efficiency of foot-based and grid-based learners in section 7. In our investigation of learning of the TS test set to which we turn to in section 3, we follow Pater et al. (2012) in choosing L-BFGS-B (Byrd et al., 1995), which has the advantage over other optimization algorithms of allowing a minimum weight of zero to be imposed (negative weights make constraints have the opposite of their intended effect). It is also generally more effective at avoiding local optima, and faster to converge. Because it is significantly more complex than Gradient Descent, we refer the interested reader to Byrd et al. (1995) for a description, and note just that the approach to hidden structure and to calculating the observed and expected probabilities are the ones explained here.

2.2. Hidden structure learning by hand and with Linear Programming

Much analytic work in OT is done by hand, and it is also possible to find correct constraint weights for HG by hand, especially for relatively simple cases (see section 3.2 for one example). In section 4, we discuss and apply an automated method for finding constraint weights that uses Linear Programming (LP). A correct constraint weighting in these cases gives a correct candidate in each tableau highest Harmony, which entails that it has the highest probability if the MaxEnt probability calculation were applied. The constraint weights tend to be much lower than in the MaxEnt analyses presented below, because they are chosen to minimize the summed weights of the constraints. One could make the probabilities of the correct candidates be arbitrarily close to 1 in MaxEnt by scaling the LP weights by a constant of sufficient magnitude. See further Pater (2016, sec. 5) and Anttila and Magri (2018) on the relationship between deterministic HG and MaxEnt.

3. The TS test set with foot-based constraints

3.1. Methods and Results

The TS test set consists of 124 languages that can be generated by a ranking of the 12 constraints in (7). The statements are our own, using the “Assign a violation...” schema of McCarthy (2008). As mentioned in the introduction, this constraint set is based on

proposals in Prince and Smolensky (1993/2004) and McCarthy and Prince (1993); we discuss alternative formulations of some of the constraints in sec. 4.

(7) Constraints from the TS Test Set

- *Nonfin*: Assign a violation if the final syllable of a word is footed.
- *WordFoot-R*: Assign a violation if the final syllable of a word is not footed.
- *WordFoot-L*: Assign a violation if the initial syllable of a word is not footed.
- *FtBin*: Assign a violation for each foot that consists only of a light syllable.
- *Iambic*: Assign a violation for each foot if its final syllable is not stressed (no violation for monosyllabic feet).
- *FootNonfin*: Assign a violation for each foot if its final syllable is stressed (every monosyllabic foot gets one violation).
- *Parse*: Assign a violation for each syllable that is not footed.
- *WSP*: Assign a violation for every heavy syllable that is not stressed.
- *Main-R*: Assign a violation for every syllable between the right edge of the word and the right edge of the foot that has primary stress.
- *Main-L*: Assign a violation for every syllable between the left edge of the word and the left edge of the foot that has primary stress.
- *AllFeet-R*: Assign a violation for every syllable between the right edge of the word and the right edge of each foot.
- *AllFeet-L*: Assign a violation for every syllable between the left edge of the word and the left edge of each foot

There are 62 tableaux, one for every combination of light and heavy syllables for words two syllables through five syllables in length, and one each for six and seven syllable strings of light syllables. Candidates are all the possible footings of those strings into maximally binary feet, with one of the feet being the primary stress. The learning datum for each tableau is a stress pattern over the string of light and heavy syllables, that is, an indication of the correct placement for the primary stress and any secondary stresses. An example input file can be found in the Supplementary Materials at [this URL](#).

For each of the 124 languages we ran our L-BFGS-B learner with all weights starting at 1, and also 10 times with each weight randomly sampled from a uniform distribution from 0 to 10. There was no regularization applied in the objective function.¹ We applied two success criteria, which turned out to be equivalent in that they agreed on whether each run was successful (see Pater, 2014, on success criteria for a probabilistic model with this type of learning problem). In one, a learning run was successful if in every tableau the candidate with the highest probability was consistent with the learning datum. In the other, a learning run was successful if the summed probability of the consistent candidates was > 0.90 . These learning experiments can be replicated using the software and input files provided in Prickett and Pater (2024).

When the learner is initialized with constraint weights set to 1, 115/124 of the languages are learned successfully (93%). This success rate is just slightly lower than

¹We experimented with various regularization settings to see if they would result in more successful learning of difficult languages. They did not.

Jarosz’s (2015) best results of 95.73%, which used Expectation Driven Learning and a pairwise ranking grammar.

When the initial weights were randomly sampled, the learner was successful 91% of the time across 10 runs of each of the languages. The randomly initialized learner was successful in at least one of the ten runs for 121/124 (98%) of the languages. Two of the languages that were never successfully learned with random initialization were learned successfully with initialization at 1, leaving just one language that our learner never successfully learned (labeled language 79 in the test set).

Based on these results, we identified 11 languages as “Hard”: the nine languages that initialization at 1 failed on, and two more that random initialization never succeeded on. Of those 11 languages, all of which seem unattested, 8 are cases in which a correct analysis must use iambic feet with nonfinality to place primary stress on the penultimate syllable. The other 3 languages are ones that have complex patterns of quantity sensitive stress placement in which a correct analysis uses a mix of trochaic and iambic feet. Language 79, the hardest of the languages, is in this second group.

We were able, with some effort, to construct an analysis by hand for language 79, but it remains unclear what distinguishes it and the other two languages in this group from some other languages in the TS test set that our learners find relatively easy.² We therefore put these aside and focus on the iambic nonfinality languages.

3.2. *Iambic Nonfinality and learnability*

One of the languages that our learner tended to fail on, language 52 in the test set, has a particularly simple description: stress the final syllable of a bisyllabic word with a light initial syllable, else stress the penultimate syllable. It also falls out straightforwardly from a ranking or weighting of the constraints we are using, and of similar ones from the OT stress typology literature. The version of the TS test set we used (from Bruce Tesar *via* Paul Boersma) does not come with correct rankings or descriptions of the languages, so the descriptions and analyses presented here are our own.

²We were also able to find the same analysis for language 79, as well as two others, using the Linear Programming method (LP) described in sec. 4. The constraint weights in these analyses are strikingly high compared to the ones found with LP that we present in that section (the LP method finds an analysis with the lowest possible sum of weights). That such high required weights would make learning difficult is intuitively plausible, and deserves more systematic study. See relatedly the discussion of depth of constraint ordering in the next section.

We start with a set of weights that we found by hand that yields a correct analysis, that is, one in which a candidate that is consistent with the target learning datum has the highest probability (=highest Harmony) for all of the tableaux. The constraints with non-zero weights are shown in (8), alongside the parse of the bisyllabic word types and one trisyllable. Final stress in the first two example parses violates Nonfin, but having initial stress without violating that constraint would require a violation of FtBin. A heavy syllable can form a foot without violating FtBin, as it does in examples with initial heavies, which also satisfy Nonfin. The last example shows the binary nonfinal iambic foot that leads to penultimate stress in an [L L L] word. Iambic is unviolated in this analysis and could also have an arbitrarily higher weight; Parse is violated in order to satisfy Nonfin and must therefore have a lower weight than that constraint. The analysis will fail if either of these constraints has zero weight (e.g., by failing to choose iambs over trochees).

(8) Weights and parses for correct analysis of language 52 found by hand

<i>Constraint Name</i>	<i>Weight</i>	<i>Example Parses</i>
FtBin	15	(L L1)
Nonfin	10	(L H1)
AllFeet-R	5	(H1) L
Parse	1	(H1) H
Iambic	1	(L L1) L

This pattern of interaction where a lone light syllable is footed with an adjacent one is the iambic version of the analysis that Prince and Smolensky (1993/2004) provide for Latin, a trochaic language that usually leaves final syllables unparsed. In their analysis, a word with an [L H] syllabic shape is footed [(L1 H)] (see further section 4.1 below). Suspension of nonfinality in small words is also possible in parametric theories with extrametricality (e.g. Hayes, 1995). It is therefore hard to know how it would be ruled out in a foot-based theory of stress typology.

We have been unable to find an example of a language with this pattern of stress in the typological literature that we have surveyed, and no phonologists that we have asked about it could think of one. The only prior discussion of a similar typological gap that we have found is that of Kager (2005, p. 24), who notes that a pattern in which only disyllables have final stress and all other word types have penultimate is produced by a standard set of Gradient Alignment constraints (McCarthy and Prince, 1993), but not by Rhythmic Licensing Theory (Kager, 2001), which is a grid-based theory of stress placement. He classifies it as unattested. Kager's typological investigation is limited to quantity insensitive patterns, so the exact pattern here is not considered.

Our learner, with a random initialization, found the weights in (9), rounded to one decimal point, which yield the same parses as shown in (8). In this analysis, Nonfin has the highest weight, and the fact that [(L L1)] and [(L H1)] have higher probability than [(L1)L] and [(L1)H] respectively comes from a gang effect. FtBin, AllFeet-R, Main-R, WordFoot-R and Parse all prefer final stress for both [L L] and [L H], as does WSP for [L H], and their summed weight is greater than that of Nonfin. For this and for all other analyses discussed in this paper, a file showing the highest probability candidate for every tableau, and their constraint violations, can be found in the Supplementary Materials at [this URL](#).

- (9) Correct weights for language 52 found with a batch MaxEnt learner (random init)

<i>Constraint Name</i>	<i>Weight</i>
Nonfin	29.3
FtBin	17.3
Iambic	16.1
AllFeet-R	5
Main-R	5.1
AllFeet-L	3.4
WordFoot-R	2.6
FootNonfin	2.2
Parse	1.3
Main-L	0.9
WSP	0.8
WordFoot-L	0.0

Random initialization led to successful learning only one out of ten tries for this language, and initialization at 1 also failed. The final weights from initialization at 1 are shown in (10), along with the parses to which they assign highest probability for the five word types also shown in (8). The asterisked word types have stress in the wrong place relative to the learning data. This is a trochaic analysis, with FootNonfin, the TS trochee constraint, having higher weight than Iambic, and constraints wanting a foot at the right edge having higher weight than Nonfin, which had zero weight.

- (10) Incorrect weights for language 52 found with a batch MaxEnt learner (init 1)

<i>Constraint Name</i>	<i>Weight</i>	<i>Example Parses</i>
AllFeet-R	8.5	*(L1 L)
WordFoot-R	6.3	*(L1 H)
FootNonfin	5.6	(H1 L)
Main-R	4.4	(H1 H)
AllFeet-L	2.2	(L 1L)
Iambic	1.9	
Main-L	1.8	
WSP	1.2	
Parse	1.1	
WordFoot-L	0.0	
FtBin	0.0	
Nonfin	0.0	

This pattern has been previously discussed in the learning literature. Jarosz (2013, fn. 10) notes that none of the OT learners she tested on it using the TS candidate and constraint sets found a correct analysis, and speculates that the source of the difficulty is that “learners can be led astray by an overwhelming amount of evidence consistent with a preferred (but incorrect) analysis”. A trochaic analysis works for all word types except the light-initial disyllables (these cannot be parsed with final stress without violating the TS version of the Trochee constraint, FootNonfin), that is for all but 2 of the 62 tableaux in this test set.

For the learners to be winding up so often in this local optimum, there must be something that prefers the trochaic analysis of penultimate stress over the iambic one. One likely factor is the greater depth of constraint ordering in the iambic analysis, which requires an active AllFeet-R constraint that is itself overridden by Nonfin. The placement of the foot in the trochaic analysis is supported by several constraints: AllFeet-R, WordFoot-R, and Main-R, all of which have relatively high weight in (10). In MaxEnt, the iambic analysis requires AllFeet-R to have sufficient weight on its own to rule out feet further to the left, and Nonfin must have sufficiently higher weight than AllFeet-R to keep the foot off the final syllable. The trochaic analysis allows a group of lower weighted constraints to jointly place the foot in final position. This seems likely to structure the learning space in such a way that it is easier to wind up with the trochaic analysis, and harder to abandon it.

The other languages with iambic nonfinality amongst our “hard” languages are 41, 44, 45, 56, 119 and 121 in the test set. These have more data that are inconsistent with a trochaic analysis, and were generally easier to learn in our experiments. The proportions of successful runs over all 11 batch MaxEnt runs per language (initialization at 1, and 10 random) are provided in (11). The last two rows provide the average for these 7 languages, and for the 124 in the test set.

(11) Success rates for iambic nonfinal languages

Language	Success rate
41	0.73
44	0.36
45	0.45
52	0.09
56	0.45
119	0.18
121	0.18
All 7	0.35
All 124	0.90

Most of the languages in which more data was inconsistent with a trochaic analysis (languages 41, 44, 45, 56) were more successfully learned than the two languages in which only two of the 62 learning data were inconsistent with trochaicity (language 121 along with language 52). This supports Jarosz’s (2013) suggestion that language 52 is hard because of this small amount of data inconsistent with the incorrect trochaic analysis. We note though that language 119 was no easier than 121 for the batch MaxEnt learner even though less of the data for 119 is consistent with a trochaic analysis; we have no explanation for why this was so.

Boersma (2003) first brought up the possibility that the failure of learners on parts of the TS test set might help to explain typological gaps, if those languages were unattested. This possibility has remained unexplored until now, but the initial results presented here seem promising. This case extends earlier demonstrations that learning might shape stress typology (Bane and Riggle, 2008; Heinz, 2009; Staubs, 2014; Stanton, 2016) to cases of ambiguity caused by hidden structure. Further work is needed to show that learning will in fact have the desired effect on typology, for

example through agent-based simulations (Staub, 2014). It also remains possible that the true source of the gaps is in the grammatical theory; these systems are not produced by metrical theories that place stress without reference to feet (Prince, 1983; Bailey, 1995; Gordon, 2002; Kager, 2005).

4. Representing and learning attested patterns of stress

In the last section we showed that a MaxEnt learner equipped with a means of dealing with hidden structure succeeds on the vast majority of the languages in the TS test set, and that its failures may help to explain typological gaps. This success, and also the potentially useful failure, on a well-studied test set provides support for the project of using the MaxEnt framework for a learning comparison of grammatical theories.

As mentioned in the introduction, the “attested typology” that we consider here is a set of 61 patterns provided in StressTyp2 (Goedemans et al., 2015) as finite state transducers (FSTs).³ A list of the 61 patterns is provided in the Appendix. Descriptions of the patterns, and the sources on which those descriptions are based, can be found in StressTyp2 by entering the pattern number provided in the Appendix. We have not taken on a critical examination of whether the patterns in this database are in fact accurate representations of the languages that are cited as instantiations of these patterns, nor have we attempted to make this typology more representative of the full range of cross-linguistic variation by adding to it. Both of these are important steps in future research. We chose to put them aside so as to be able to produce a set of initial results for our proposed approach to comparing grammatical theories on the basis of their learnability.

The TS test set is designed to be representable by the constraint set that they adopt. When studying learning of attested typology, it can happen that we don’t know in advance which of the patterns a given constraint set can represent. An observed learning failure could thus have two sources: either the language can simply not be represented, or it can be represented, but the learner failed to find a correct grammar.

To address this issue, we implemented the Linear Programming (LP) method for determining whether a pattern can be represented by a given set of weighted constraints presented in Potts et al. (2010). The existing implementation in OT-Help (Staub et al., 2010) was not sufficient for our purposes. When there is hidden structure, the only way of determining in OT-Help whether a given pattern can be represented is by generating a full predicted typology, and then checking if the pattern in question is contained within it. And OT-Help cannot calculate full predicted typologies for tableau sets as large as ours.

This new LP implementation, SoftStress (Lee and Pater, 2024) uses the general approach to typology calculation from OT-Help, but adds the option of limiting the search space to those full representations (e.g. prosodifications) that are consistent with a given set of overt structures (e.g. stress patterns). It also improves on OT-Help

³StressTyp2 in fact provides FSTs for 107 different stress patterns and out of these, there are 61 that are in principle representable by the constraint sets we consider here. For example, the constraint sets have no means to represent lexical conditioning of stress patterns (e.g. exceptionality) and ternary weight systems. Additionally, some pairs of patterns only differed in whether they imposed word size minima (i.e. bans on monosyllabic or monomoraic words), which we abstract from here.

in terms of the size of tableau sets it can calculate a typology for, and the speed with which it calculates a typology.

The LP method works with a deterministic version of Harmonic Grammar (see [Pater 2016](#)), which picks one of the candidates in a tableau as optimal if it has a sum of weighted violation scores that is higher than any of its competitors (=higher Harmony). The set of languages that can be represented in deterministic HG is identical to the set of languages that give the optima in those languages highest probability in their candidate sets in MaxEnt ([Pater 2016](#), sec. 5). The LP method indicates whether or not a set of candidates can be made to each have the highest Harmony in its tableau, and also provides a set of weights if they can. The weights are chosen to minimize their summed values.

We also ran learning simulations using Gradient Descent. We chose this algorithm over L-BFGS-B to more easily compare the learning trajectories of each model. All learners used a learning rate of 4, initial weights of 1, and were run up to 10,000 epochs or until the grammar assigned a probability to every correct stress pattern above 0.9, where the probability of a pattern is the sum of the probabilities of the prosodifications that yield that pattern. As in our L-BFGS-B experiments, we used no regularization. SoftStress implements Gradient Descent as well as the LP solving method, and also provides input files to replicate all of the computational experiments in this section.

By studying learning and representation of the attested typology using MaxEnt learning and our LP method, we made some discoveries about the choices of constraint formulation in the TS constraint set that we believe are novel. Before turning to the typology and learning comparison with grid-based theories, we present those results, which led to our inclusion of a revised foot-based constraint set in that comparison.

4.1. Latin and a revised foot-based constraint set

The TS constraint set cannot represent the stress pattern of Latin, which is in the StressTyp2 database, and which is analyzed with constraints similar, but not identical, to those in TS by [Prince and Smolensky \(1993/2004\)](#). The prose description in StressTyp2 is as follows:

“In words of three or more syllables, primary stress falls on the penultimate syllable if it is heavy, else on the antepenultimate syllable... In words of two or fewer syllables, primary stress falls on the initial syllable. In words of all sizes, there is no secondary stress.”

The StressTyp2 description also includes the absence of light monosyllabic words. Following TS, we abstract from restrictions on word size.

For the constraint demanding trochaic feet, TS choose a version that is only satisfied by a bisyllabic left-headed foot, citing work on asymmetries between iambic and trochaic feet (see the review in [Hyde, 2011](#)). This departs from [Prince and Smolensky \(1993/2004\)](#), whose analysis of Latin includes monosyllabic trochaic feet. [McCarthy and Prince \(1993\)](#) have a FootForm (Trochaic) constraint that appears to admit only bisyllabic feet, but their footnoted Alignment-based formulation requires only that the head be at the left edge, which is satisfied by a monosyllable. The TS main stress placement constraint is also a departure from [Prince and Smolensky \(1993/2004\)](#), who

use an Edgmemost constraint that counts the number of syllables between the main stress and the edge of the word. The TS constraint measures instead the number of syllables between the edge of the foot that bears main stress and the word edge, which has a precedent in the Align-Head constraint of McCarthy and Prince (1993).

The problem for an analysis of Latin with the TS constraint set is that nothing prefers penultimate stress over antepenultimate stress when both syllables are heavy, so long as feet are trochaic. Parses producing antepenultimate and penultimate stress are shown in (12); X stands for a syllable that could be heavy or light. Both of these parses equally violate WSP, since each has a single unstressed heavy syllable. Antepenultimate stress (12a.) does better on Parse-Syl and FootNonfin (as well as the left edge alignment constraints). The only constraint in the TS constraint set that prefers the desired (12b.) is Iamb, since the monosyllabic foot satisfies it. But if that constraint has sufficient weight relative to other constraints to choose (12b.), other strings would be misparsed: in particular, a [L L X] string would receive penultimate stress.

(12) Parses for [H1 H X] string

- a. (H1 H) X
- b. H (H1) X

If FootNonfin is replaced by a Trochee constraint that penalizes feet with stressless initial syllables, then (12b.) no longer violates it. And if the main stress constraint assigns violations by counting syllables between the right edge of the word and the syllable bearing main stress, rather than the foot bearing main stress as in TS's Main-R, it prefers (12b.) over (12a.). In addition, Iamb can act in a gang effect with Trochee to choose (12b.) since that representation does not violate Trochee.

Our revised constraint set (TS-Rev) is the same as that of TS presented in (7) above (TS-OG), except that FootNon-fin is replaced by Trochee (13a.), and Main-R and Main-L are replaced by Main-Syl-R (13b.) and Main-Syl-L (13c.).

(13) Revised foot-based constraints

- a. *Trochee*: Assign a violation for each foot if its initial syllable is not stressed
- b. *Main-Syl-R*: Assign a violation for every syllable between the right edge of the word and the syllable that bears main stress
- c. *Main-Syl-L*: Assign a violation for every syllable between the left edge of the word and the syllable that bears main stress

Another difference between the TS constraint set and the one used by Prince and Smolensky (1993/2004) in their analysis of Latin is in the formulation of the Nonfinality constraint. Nonfinality in Prince and Smolensky (1993/2004) assigns one violation to a stressless final syllable that is parsed into a main stress foot but is not itself stressed, and two violations to a final syllable that receives main stress (and is footed). The TS constraint assigns a single violation to all footed final syllables. We examined the effect of adding an additional Nonfinality constraint to our revised foot-based constraint set that penalizes final stressed syllables (14). We call this constraint set TS-Rev-Nonfin.

(14) Additional foot-based constraint

- Nonfin-main*: Assign a violation to a final syllable that has primary stress

We used the StressTyp2 FST implementation of Latin to generate the stress patterns of all combinations of heavy and light syllables for strings 2-5 syllables in length, and for strings of light syllables 6-7 syllables in length. We then used these as learning data with the same candidates as for the TS test set described above: all possible parses of the syllable strings, with one foot per word that is the head of the word (main stress), and any number of further dependent feet (secondary stresses), where feet can be either monosyllabic or disyllabic, and right-headed or left-headed. Each of the resulting representations was assigned constraint violations for TS-OG, TS-Rev, and TS-Rev-Nonfin.

With the TS-OG constraint set, the pattern was not learned, and our LP method indicated that it was not representable by this constraint set. With TS-Rev, LP found two analyses, but learning was not successful. With TS-Rev-Nonfin, LP found four analyses: the two found using TS-Rev, plus two additional ones, and learning was successful. This is related to an important general point. It might seem obvious that limiting the hypothesis space would help learning. In this case, learning was successful only with the larger hypothesis space, even when both provided a representation of the pattern. What is crucial is not the size of the hypothesis space, but its structure. Here, the structure of the larger hypothesis space allowed the learner to avoid a local minimum.

The constraint weights that our LP method found for the two analyses for Latin with the TS-Rev constraint set are shown in (15); constraints with zero weight are omitted. These are weights that are correct for deterministic HG: the Harmony (weighted sum of violations) is highest for a parse that corresponds to the correct stressing of every string of syllables. They would also assign highest probability to those parses in MaxEnt, but for those probabilities to approach 1, they would have to be scaled (all multiplied by a large enough number).

(15) Weights for the two TS-Rev analyses found with Linear Programming

Solution 1		Solution 2	
<i>Constraint Name</i>	<i>Weight</i>	<i>Constraint Name</i>	<i>Weight</i>
Nonfin	7	Nonfin	5
Trochee	4	Trochee	2
FtBin	4	FtBin	2
Iamb	2	Main-Syl-R	1
Main-Syl-R	1	AllFeet-R	1
AllFeet-R	1		

The analyses are in many ways similar, and we will start with the similarities. In both analyses Nonfin has the highest weight, and in both of them the final syllable is always unfooted in the optimal forms. This corresponds to the familiar extrametricality-based analysis proposed by Hayes (1980, p. 240).⁴ Final non-footing creates violations of FtBin in the optimal forms in (16). As we will discuss shortly, the Prince and Smolensky analysis has bisyllabic trochaic parses for these cases. Here those bisyllabic parses have lower harmony than the monosyllabic feet because Nonfin has a higher weight than the

⁴In Hayes' analysis of English nouns with stressless final syllables ("[t]he situation here is clearly the same as in Latin"), the final syllable is made extrametrical, but is footed by stray adjunction later in the derivation.

sum of weights of FtBin and AllFeet-R, both of which are violated in these optimal forms.

(16) (L1) L (L1) H

FtBin is active in choosing the optimum in longer words with light penults, which get stress on the antepenult, as in the example in (17). MainSyl-R would prefer penultimate stress, but has a lower weight than FtBin.

(17) (L1 L) H

Stress is on a heavy penult because of active Main-Syl-R. This parse is better than antepenultimate stress because Main-Syl-R has weight 1, and the constraints that would prefer antepenultimate stress all have zero weight.

(18) L (H1) L

The difference between the analyses is in what they do with an [H L X] word. Solution 1 parses the heavy syllable on its own, corresponding to the moraic trochee analysis of Latin (Mester, 1994), while solution 2 parses it with the light syllable, corresponding to a quantity sensitive trochee analysis (Jacobs, 2003).

(19) Solution 1: (H1) L L Solution 2: (H1 L) L

Interestingly, Solution 1 does not rely on a constraint against (HL) feet - we don't have such a constraint - and Solution 2 is not preferred by Parse, which has zero weight. These would be the constraints one might expect would be at work, based on prior analyses. Here Solution 1 prefers [(H1) L L] because it satisfies Iamb, and solution 2 prefers [(H1 L) L] because it better satisfies AllFeet-R.

Gradient Descent did not find a set of weights that yields the parsings of either of these analyses. Here are the weights after 1000 epochs, which do not grant highest probability to the correct stress pattern for all word shapes:

(20) Weights for an incorrect analysis of Latin found by Gradient Descent

<i>Constraint Name</i>	<i>Weight</i>
AllFeet-R	19.49
Nonfin	14.16
Trochee	9.67
WSP	7.31
Parse	6.62
Main-Syl-L	6.32
AllFeet-L	3.67
FtBin	3.14
Main-Syl-R	1.79
WordFoot-L	0.10

This grammar parses bisyllables into binary feet, and leaves the final syllable unfooted in some longer words. One example of final syllable non-footing, which yields the correct result in this case, is in (21).

(21) (L1 L)L

Final syllable non-footing is a gang effect, since Nonfin has a lower weight than AllFeet-R. The other participants in the gang effect are MainSyl-L and AllFeet-L, which both prefer [(L1 L) L] over [L (L1 L)]. The sum of the weights of these two constraints and Nonfin is greater than the weight of AllFeet-R.

The final syllable is also footed in some correct parsings, as in example (22). The fact that this wins over [(L1 H) L] is due to WSP ganging up with AllFeet-R to overcome the constraints that prefer antepenultimate stress.

(22) L (H1 L)

This analysis fails to place stress on a heavy penult when WSP is equally satisfied by stressing a heavy antepenult. The incorrect parse is shown in (23). Note that the incorrect grammar found by TS-Rev has the same problem as the TS-OG constraint set.

(23) *(H1 H) L

This analysis appears to be a local optimum: letting the learner run to 10,000 epochs still does not result in a correct analysis.

The constraint set with the additional Nonfinality constraint that penalizes just stressed final syllables ('Nonfin-main' in (14)), TS-Rev-Nonfin, finds two additional analyses, and allows Gradient Descent to find a correct analysis in less than 1000 epochs. Both of these solutions parse the final syllable with an initial L in bisyllables:

(24) Additional set of optimal parses for TS-Rev-Nonfin+LP

(L1 L)

(L1 H)

(H1) L

(H1) H

These parses are the same as in the analysis of Latin offered by Prince and Smolensky, who justify it based on the shortening of final H in [(L1 H)] in a variety of Latin, and also based on the absence of monomoraic words, which they analyze using an undominated FtBin constraint.

The two analyses differ along the lines discussed for the first two solutions - one has [(HL)X] and the other has [(H)LX]. The weights for the [(HL)X] solution appear in (25). As in the other [(HL)X] analysis, Solution 2 in (15), Iambic has zero weight, and AlignFeetRight has a weight of 1, which prefers the footing of the penult.

(25) Weights for TS-Rev-Nonfin+LP Solution 3

<i>Constraint Name</i>	<i>Weight</i>
Nonfin	3
FtBin	3
Trochee	2
Nonfin-main	2
Main-Syl-R	1
AllFeet-R	1

FtBin and Nonfin, which conflict on whether the final syllable is parsed for a [L X] form are tied. [(L1 X)] wins because of a gang effect, with AllFeet-R breaking the tie. The role of Nonfin-main is to keep stress off of the final syllable as in the incorrect parse *[L1 (H)], which would satisfy FtBin, and violate Nonfin equally as correct [(L1 H)].

The fourth solution, shown in (26), gives Iambic enough weight to prefer [(H)LX] over [(H L)X]

(26) Weights for TS-Rev-Nonfin+LP Solution 4

<i>Constraint Name</i>	<i>Weight</i>
FtBin	5
Trochee	4
Nonfin-main	4
Nonfin	3
Iambic	2
Main-Syl-R	1
AllFeet-R	1

With this constraint set, Gradient Descent reaches the correctness criterion after 204 epochs. The weights are shown in (27). These weights yield the same parsings as Solution 2 above, where the parsings are the highest probability candidates.

(27) Weights for TS-Rev-Nonfin+Gradient Descent

<i>Constraint Name</i>	<i>Weight</i>
Nonfin	11.95
Nonfin-main	8.90
AllFeet-R	7.98
Trochee	6.64
FtBin	5.77
WSP	4.37
Main-Syl-R	3.83
Main-Syl-L	3.59
Iamb	2.57
AllFeet-L	0.28

As we mentioned at the outset of this subsection, the Gradient Descent simulations with these two constraint sets show that an increase in size of the hypothesis space does not necessarily make learning harder; in this case the increase facilitated it. This does not mean that increasing the hypothesis space always improves learning; again, it is the structure of the space that is important. We found, for example, that when we kept the TS FootNonfin constraint alongside the Trochee constraint, local optima became more common compared to replacing it with Trochee as in our revised constraint sets (see further section 7).

4.2. *Québécois and related quantity insensitive patterns*

Besides Latin, one of the patterns that escapes the TS-OG constraint set, and which TS-Rev and TS-Rev-Nonfin can represent, is a quantity insensitive pattern in which main stress always falls on the final syllable, and secondary stress always falls on the

initial one. StressTyp2 cites “Armenian (SPD)”, “French, Canadian”, and “Hinalug” as instantiating this pattern. We will call the pattern Québécois.

Gordon (2011: 158) draws attention to the difficulties that this pattern and related ones can pose for foot-based theories in general. First, it might be hard to see what would prefer the correct parsing in (28a.) over its competitors (28b.) and (28c.). Both syllables are parsed in all of the candidates, and (28b.) and (28c.) avoid violations of Foot Binarity.

- (28) Competing parses of [X X] string
- a. [(X2)(X1)]
 - b. [(X X1)]
 - c. [(X1 X)]

Second, in a longer string, it can also be hard to see how an initial and final stress could coexist, since they seem to require an iambic foot at the right edge and a trochaic one at the left. With some constraint sets, there is nothing that would prefer the correct (29a.) over (29b.) and (29c.).

- (29) Competing parses of [X X X X] string
- a. [(X2 X)(X X1)]
 - b. [(X2 X)(X1 X)]
 - c. [(X X2)(X X1)]

The TS-OG constraint set is in fact an example of a foot-based constraint set that cannot represent this system. Learning with Gradient Descent fails to find a correct grammar, and application of the LP method finds the problem to be one of representation.

This system can be represented by TS-Rev and TS-Rev-Nonfin because MainSyl-R prefers [(X2)(X1)] over [(X1 X)], and [(X2 X)(X X1)] over [(X2 X)(X1 X)]. When using Gradient Descent with either constraint set a correct grammar is found after 104 epochs. The weights for TS-Rev are shown in (30); those for TS-Rev-Nonfin are almost identical.

- (30) Weights for TS-Rev with Gradient Descent for Québécois

<i>Constraint Name</i>	<i>Weight</i>
Main-Syl-R	12.25
Trochee	9.15
WordFoot-L	5.92
AllFeet-L	5.17
Parse	4.26
WordFoot-R	3.71
Iamb	2.23
AllFeet-R	1.76
WSP	0.43

This analysis uses trochaic feet to place stress on the initial syllable: Trochee has higher weight than Iamb. Main-Syl-R has a sufficiently high weight to make final stress optimal. For words four syllables in length or greater, the final foot is iambic in the

highest probability candidate, though the second best candidate has a monosyllabic final foot, and helps to bring probability over 0.9 for the correct stress pattern. Examples of correct parses and their probabilities appear in (31).

- (31) Correct parses and probabilities for TS-Rev with Gradient Descent for Québécois
- [(L2) (L1)] : 0.90
 - [(L2 L) (L1)] : 0.95
 - [(L2 L) (L L1)] : 0.55
 - [(L2 L) L (L1)] : 0.42

The LP method finds 15 analyses for Québécois with the TS-Rev constraint set. All have Trochee above Iamb, and all have Main-Syl-R with non-zero weight. The weights for the analysis that makes use of the smallest set of constraints is shown in (32).

- (32) Weights found for TS-Rev with LP for Québécois
- | <i>Constraint Name</i> | <i>Weight</i> |
|------------------------|---------------|
| Main-Syl-R | 5 |
| Trochee | 4 |
| AllFeet-L | 3 |
| Parse | 2 |

The parses it picks are identical to the highest probability parses in the analysis found by Gradient Descent, as in these examples that match those in (31) above:

- (33) Example parses for TS-Rev with LP for Québécois
- [(L2) (L1)]
 - [(L2 L) (L1)]
 - [(L2 L) (L L1)]

Because Main-Syl-R targets the position of a prominent syllable, rather than of a foot, the theory that incorporates it can be seen in a sense as a mixed foot- and grid-based system (though the formulation is in terms of the head of the prosodic word, rather than main stress or highest level grid mark).

Before we attempted to learn Québécois with a constraint set with Main-Syl-R, we expected that it would escape a foot-based analysis, and before attempting to learn Latin with the TS-OG constraint set we did not know that it could not be represented by it. For us at least, both of these findings were novel.

5. Typology across feet and grids

In this section, we compare the typological coverage of four constraint sets in terms of which of the 61 stress patterns from StressTyp2 they can represent: the two foot-based constraint sets that we have been calling TS-OG and TS-RevNonFin and two grid-based constraint sets. We will shorten the abbreviation of TS-RevNonFin to TS-REV since we will only use a single revised constraint set, the one that includes the additional Nonfinality constraint. It captures 2 languages that cannot be represented

without that constraint, 194 (Mongolian Khalkha SPD) and 199 (Nubian Dongolese), but otherwise the typological coverage is identical.

The first grid-based constraint set is the one proposed by [Gordon \(2002\)](#), with the addition of a Weight-to-Stress constraint. Gordon shows how this constraint set aligns with the typology of quantity insensitive systems, and [Hayes and Wilson \(2008\)](#) provide positive learning results on the attested typology with a MaxEnt learner that constructs its own constraints. As far as we know, neither typology nor learning has been previously studied with these grid-based constraints and quantity sensitive systems. The constraints are defined in (34) using the "Assign a violation..." format.

(34) Rephrased constraints from [Gordon \(2002\)](#), plus Weight-to-Stress

- *Align*(X_1 , L, 0, PrWd) (*Align1LPrWd*): Assign a violation for every grid mark at level 0 intervening between each grid mark at level 1 and the left edge of the word.
- *Align*(X_1 , R, 0, PrWd) (*Align1RPrWd*): Assign a violation for every grid mark at level 0 intervening between each grid mark at level 1 and the right edge of the word.
- *Align*(Edges, level 0, PrWd, X_1) (*AlignEdges*): Assign a violation for every grid mark on the edges at level 0 that does not have a grid mark at level 1 (maximum 2 violations per word).
- *NonFinality*: Assign one violation if the final syllable has a grid mark at level 1.
- **Clash*: Assign a violation for every pair of consecutive syllables that have a level 1 grid mark.
- **Lapse*: Assign a violation for every pair of consecutive syllables lacking a level 1 grid mark.
- **ExtendedLapse*: Assign a violation for every triplet of consecutive syllables lacking a level 1 grid mark.
- **LapseRight*: Assign one violation mark if there is more than one level 0 grid mark (≥ 2) intervening between the rightmost level 1 grid mark and the right edge of the word.
- **LapseLeft*: Assign one violation mark if there is more than one level 0 grid mark (≥ 2) intervening between the leftmost level 1 grid mark and the left edge of the word.
- **ExtendedLapseRight*: Assign one violation mark if there are more than two level 0 grid marks (≥ 3) intervening between the rightmost level 1 grid mark and the right edge of the word.
- *Align*(X_2 , L, 1, PrWd) (*Align2LPrWd*): Assign a violation for every grid mark at level 1 intervening between each grid mark at level 2 and the left edge of the word.
- *Align*(X_2 , R, 1, PrWd) (*Align2RPrWd*): Assign a violation for every grid mark at level 1 intervening between each grid mark at level 2 and the right edge of the word.
- *WSP*: Assign a violation for every heavy syllable that does not have a level 1 grid mark

We also examine a constraint set that revises the main stress placement constraint to count the number of syllables from the edge, parallel to the revised foot-based main stress constraint. The revised main stress constraints are in (35).

(35) Revised grid-based constraints

- *Align*(X_2 , L, 0, PrWd) (*Align2LPrWd_σ*): Assign a violation for every grid mark at level 0 intervening between each grid mark at level 2 and the left edge of the word.
- *Align*(X_2 , R, 0, PrWd) (*Align2RPrWd_σ*): Assign a violation for every grid mark at level 0 intervening between each grid mark at level 2 and the right edge of the word.

The Venn diagram in Figure 1 conveniently represents the overall coverage of the four theories. The numbers are those assigned in StressTyp2 to the patterns; see the Appendix for a full list. 129 is Québécois, which can be represented by TS-REV and by the grid-based theories, but not by TS-OG, and 178 is Latin, which can only be represented by TS-REV.

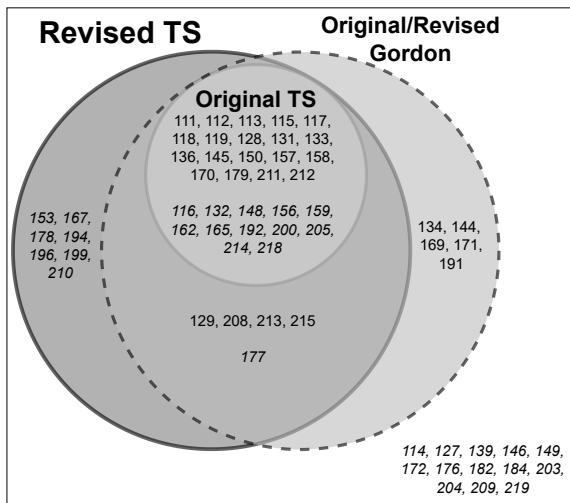


Figure 1. Patterns represented by four theories of stress. The numbers in italics are quantity sensitive patterns; others are quantity insensitive.

The innermost region shows the set of 31 languages that the TS constraints can represent. The revised TS constraint adds 12 more languages, while the Gordon constraint sets add 10. There is no difference between the original and revised Gordon constraint sets in terms of which of these patterns they can represent.

Of the 5 languages, including Québécois, that TS-REV and the grid-based theory capture and TS-OG does not, 4 are quantity insensitive (QI). The discussion of Québécois above generalizes to the three other QI systems (see also [Gordon, 2011](#), 158 for

discussion of these patterns). 208 Shosone Gosiute and 213 Tauya have the mirror image of the Québécois stress pattern for bisyllabic words: initial primary, final secondary. This cannot be represented in TS-OG for reasons parallel to those discussed for Québécois, and as in Québécois, these can be represented in TS-REV due to change in the main stress placement constraint. 215 Udihe is the same pattern as Québécois except that bisyllabic words lack initial stress. The problem discussed above for the longer words in Québécois remains an issue here for TS-OG.

All of the 7 languages that TS-REV alone can represent are QS: along with 178 Latin, these are 153 English (SPD), 167 Hopi, 194 Mongolian, Khalkha (SPD), 196 Mongolian, Khalkha, 199 Nubian, Dongolese and 210 Sindhi (SPD). The issue discussed for Latin for TS-OG applies to all of them. In each case, stress falls on a heavy syllable nearest to one edge (putting aside the final syllable in Latin, and sometimes in 194 Khalkha), else on a syllable further away from the edge. The problem for TS-OG is that with iambs as well as trochees, nothing militates against having a bisyllabic (H H) foot at the edge instead of a (H) foot. This also applies to 177 Kuuku-Ya?u (SPD) which has a default-to-opposite QS pattern. As in Latin, this issue is resolved for all of these languages by having main stress alignment constraints that target the main stressed syllable instead of the foot bearing main stress. The fact that these languages also escape the Gordon grid-based constraint set indicates that Weight-to-Stress alone is not a sufficient account of quantity sensitivity in that framework, just as it isn't a sufficient account in the foot-based approach. One possible amendment to the Gordon constraint set would be to include constraints that measure lapses and clashes over moras.

The five languages that only the Gordon grid-based constraint sets can represent are 134 Bhojpuri (SPD), 144 Cayubaba, 169 Icuá Tupi, 171 Ioway-Oto, and 191 Mingrelian. These are all QI languages that have some ternary stretches: either antepenultimate stress, or a ternary alternation. Even though the foot-based constraint set can account for some cases of antepenultimate stress (e.g. when the penult is light in Latin), the foundational OT constraint sets for stress are known to not provide a general account of ternarity; see [Martínez-Paricio and Kager \(2015\)](#) for recent discussion.

Finally, the 14 languages that none of the theories can represent are all QS. We haven't found any further generalizations about the types of systems in this group.

6. Learnability of the attested languages with feet

To assess the learnability of the attested languages that the foot-based theories can represent, we use the same learning algorithm as for the study of the TS languages, L-BFGS-B. This algorithm is faster to converge, and less prone to local minima than Gradient Descent. For TS-OG, the results are very similar to those obtained with the TS test set. For the attested languages, $29/31 = 94\%$ were learned successfully with initialization at 1, compared to $115/124 = 93\%$ for the TS test set. With random initialization, 10 times for each language, overall 90% of runs were successful, compared with 91% for the TS set.

For TS-REV, $42/43 = 98\%$ of the languages were learned successfully with initialization at 1, and random initialization had an overall success rate of 90%. None of

the languages failed to be learned in all runs, as happened with language 79 in the TS test set, and also language 200 Nyawaygi for TS-OG (TS-REV learned that quantity sensitive language in all 11 runs).

Language 157 Garawa was the language that L-BFGS-B with initialization at 1 failed to learn with the TS-REV constraint set. Only 4 out of 10 of the random initializations were successful. In contrast, TS-OG was successful in all 11 runs. The difference stems from the formulation of the constraints demanding left-headed feet.

In (36) the correct analysis found by the TS-OG learner initialized at 1 is shown. The foot form constraint demanding that the final syllable be stressless has the highest weight and feet are uniformly left-headed and bisyllabic. The rest of the constraints place the main stress foot at the left edge, and secondary stress feet aligned to right. Heavy syllables and light syllables are treated the same, as they should be in this quantity insensitive language.

- (36) Weights and parses for correct analysis of 157 Garawa, TS-OG, L-BFGS-B, Init 1

<i>Constraint Name</i>	<i>Weight</i>	<i>Example Parses</i>
FootNonfin	26.87	(X1 X)
MainLeft	12.54	(X1 X) X
WordFoot-L	12.00	(X1 X) X (X2 X)
Parse	11.27	(X1 X) X (X2 X) (X2 X)
AllFeet-R	6.90	
WordFoot-R	5.10	
FtBin	4.93	

We used our LP implementation to determine if this same analysis could be generated by the TS-REV constraint set, that is, whether the same set of parsings could be made to have the highest Harmony. It turns out that it cannot. The problem is that there is no constraint that would penalize the parsing of single heavy syllables, for example in a string like [(X1 X) H (X2 X)], which would therefore be harmonically bounded by the incorrect *[(X1 X)(H2)(X2 X)].

The incorrect analysis found with the TS-REV constraints with initialization at 1 is shown in (37). It achieves some quantity insensitivity, placing a binary foot at the left edge, and at the right edge of words four syllables in length and higher. However, in trisyllables, a final heavy syllable is incorrectly stressed. That monosyllabic foot would violate the TS-OG FootNonfin constraint, but satisfies Trochee. This analysis also fails to place a word-internal foot in 6 and 7 syllable words.

- (37) Weights and parses for incorrect analysis of 157 Garawa, TS-REV, L-BFGS-B, Init 1

<i>Constraint Name</i>	<i>Weight</i>	<i>Example Parses</i>
Trochee	25.39	(L1 H)
FtBin	21.51	(L1 L) L
Main-Syl-Left	17.51	*(L1 L) (H2)
Parse	16.47	(L1 L)(L2 H)
AllFeet-R	16.13	*(L1 L) L L (L2 L)
WordFoot-L	14.83	*(L1 L) L L L (L2 L)
Nonfin	13.07	
Nonfin-main	1.30	
AllFeet-L	1.24	
WordFoot-R	0.43	

A correct quantity insensitive analysis found with the TS-REV constraint set with random initialization appears in (38). The high weight of Nonfin leads to final syllables being consistently unparsed, and MainStressLeft places main stress on the initial syllable. The correct placement of other stresses is achieved with iambic footing. This is an iambic nonfinality analysis, and like the unattested patterns discussed in section 3.2, it is difficult to learn, presumably for similar reasons: an incorrect trochaic analysis covers much of the data, and the learner is biased to adopt it by the structure of the constraint set.

- (38) Weights and parses for correct analysis of 157 Garawa, TS-REV, L-BFGS-B, Random Init

<i>Constraint Name</i>	<i>Weight</i>	<i>Example Parses</i>
Nonfin	55.06	(X1) X
Parse	35.90	(X1 X) X
Main-Syl-Left	30.70	(X1) (X X2) X
Iambic	12.99	(X1) (X X2) (X X2) X
AllFeet-R	7.83	(X1 X) (X X2) (X X2) X
WordFoot-L	7.43	
AllFeet-L	6.96	
Nonfin-main	5.16	
FtBin	0.57	
WordFoot-R	0.01	

Even though TS-REV can represent Garawa, from the perspective of learning that language, the constraint set is not as successful as the TS-OG one. The TS-OG FootNonfin constraint appears to aid in the learning of quantity insensitive languages, while we have seen that the TS-REV Trochee constraint is superior in representing Latin and other quantity sensitive languages. In the next section, we will see further evidence of the TS-OG constraint set outperforming TS-REV in learning QI languages that both can represent, and TS-REV being at an advantage for QS languages.

7. Speed of learning and biases across grammatical models

In this section, we use Gradient Descent to measure the speed of learning across constraint sets. As in the Gradient Descent simulations reported in section 4, the

constraints are initialized at 1, the learning rate is set to 4, and learning is stopped when probability 0.9 is assigned to the correct stress pattern in every tableau, or when 10,000 epochs are reached.

In (39), we report success rates for the foot-based constraint sets. As expected, these are somewhat lower than for the L-BFGS-B simulations with initialization at 1 reported in the last section. For the TS-OG constraint set, the same two QS languages fail to be learned (156 Fijian and 200 Nyawaygi), and in addition, one of the QI languages is not learned (212 Southern Paiute). For TS-REV, along with 157 Garawa, three more QI languages fail to be learned: 117 Lenakel, 158 Georgian and 170 Indonesian, as well as QS Mongolian Khalka. The success rates shown in table 39 fit with the pattern observed in section 6: TS-OG outperforming TS-REV on QI languages, and TS-REV doing better on QS.

(39) Success rates with Gradient Descent for foot-based constraint sets

Constraint Set	QI	QS	All
TS-OG	0.95 (18/19)	0.83 (10/12)	0.90 (28/31)
TS-REV	0.83 (19/23)	0.95 (19/20)	0.88 (38/43)

To study the biases of the constraint sets within the languages for which learning was successful, we selected the 24 languages that were learned successfully within 10,000 epochs by all four of the constraint sets we examined (TS-OG, TS-REV, Gordon-OG, Gordon-REV). 14 were QI and 10 were QS. We then compared the number of epochs that each model needed to reach a correct analysis. Summary statistics are shown in (40).

(40) Epochs to success for 24 languages

Constraint Set	Mean	Median	SD	Min	Max
TS-OG	247.46	30.5	360.26	1	1273
TS-REV	255.17	59.5	350.81	1	1018
Gordon-OG	299.13	162.5	331.85	1	1089
Gordon-REV	400.42	164	890.73	1	4417

We refrain from concluding on the basis of these comparisons that foot-based learning is generally more efficient than grid-based learning, since in constructing this set of languages, we included only those languages that were learned successfully by the foot-based learners, taking out the hardest languages from just those constraint sets. Nonetheless, we can say that we have failed to find evidence for the superiority of the grid-based constraint sets in terms of the number of epochs to success on these 24 languages.⁵

A more valid comparison is of the biases within the constraint sets, which we turn to next.

In (41), we provide the epochs to success for the foot-based constraints separately for QI and QS languages. The speed of learning results match the pattern that we saw in rates of success: QI languages are learned more quickly by TS-OG, and QS languages are learned more quickly by TS-REV.

⁵In terms of overall compute time, grid-based constraints are a clear winner. Our learning experiments took much longer with the foot-based constraints, because each update takes longer.

- (41) Epochs to success for foot-based constraint sets broken down by quantity insensitive and quantity sensitive languages

	Constraint Set	Mean	Median	SD	Min	Max
QI	TS-OG	117.57	21.5	175.59	1	475
	TS-REV	199.86	32	305.86	1	803
QS	TS-OG	429.30	244	473.45	13	1273
	TS-REV	332.60	76.5	409.82	13	1018

We have already discussed in some detail the differences in the formulation of the constraints demanding left-headed feet, which is satisfied only by bisyllables in TS-OG's FootNonfin, and also by monosyllables in TS-REV's Trochee. While Trochee might have seemed to be a clear winner based on its greater success in representing and learning QS languages, we also have considerable evidence from learning QI languages that Trochee is not completely adequate.

One possible solution is to expand the TS-REV constraint set to include a constraint that requires bisyllabic feet. As an initial exploration of this possibility, we compared learning of Garawa and Latin for TS-REV versus TS-REV plus the FootNonfin constraint. The results are presented in the table in (42). For Garawa, the addition of the constraint does make L-BFGS-B with initialization at 1 successful, and increases the number of successful L-BFGS-B runs with random initialization from 2 to 10 (our prior test without FootNonfin reported above got 4/10). It also results in success with Gradient Descent (GD) at initialization of 1, in a relatively speedy 163 epochs. For Latin, the addition of the constraint reduces the success rate with random initialization slightly, from 9 runs to 7 (we got these same success rates in an earlier test, mentioned above). In terms of speed of learning with Gradient Descent using initialization at 1, the addition of the constraint has a slightly positive impact. If this pattern of results holds up in further testing, it would seem that the slight increase in local minima in quantity sensitive learning is well compensated for by the larger decrease in local minima in quantity insensitive languages.

- (42) Consequences of adding FootNonfin (FNF) to the TS-REV constraint set

	Constraint Set	L-BFGS-B (init 1)	L-BFGS-B (random init)	GD (init 1)
Garawa	TS-REV	Failure	2/10	Failure
	TS-REV + FNF	Success	10/10	163
Latin	TS-REV	Success	9/10	205
	TS-REV + FNF	Success	7/10	184

The table in (43) shows that while the QI languages are learned faster with the Gordon-REV constraints than the Gordon-OG constraints, the QS languages are learned faster with the Gordon-OG constraints. While the median is slightly lower for QS languages with Gordon-REV than Gordon-OG, all other measures indicate that they are learned faster with the Gordon-OG constraints. Even though the representational capacity of the two constraint sets is identical in terms of the space of patterns we have examined, there is a difference between the two in terms of the rate of learning, as a function of whether the languages are quantity-sensitive.

- (43) Epochs to success for grid-based constraint sets broken down by quantity insensitive and quantity sensitive languages

	Constraint Set	Mean	Median	SD	Min	Max
QI	Gordon-OG	204.21	103	265.61	1	875
	Gordon-REV	143.36	94	185.50	1	685
QS	Gordon-OG	432.00	303	381.96	22	1089
	Gordon-REV	760.30	274	1315.74	76	4417

For the grid constraints, there are likely two reasons that Gordon-REV is faster than Gordon-OG on the QI languages (see also [Lee et al. \(2023\)](#) for a similar finding on a slightly larger set of QI languages). First, there is the simple fact that a count of intervening syllables will usually be greater, and will never be smaller, than a count of intervening secondary stresses, which will result in a generally larger update of the main stress constraint that counts intervening syllables. Second, as Gaja Jarosz (p.c.) points out, the extent to which the weight of the main stress constraint that counts intervening level 1 grid marks (=secondary stresses) is changed depends on how probable candidates with intervening secondary stress are; this does not affect the syllable counting constraint. For QS languages, the syllable counting constraint may be too powerful in its preference for stress at an edge when main stress is not a fixed distance from the edge, as is often the case in QS languages. This could result in stages of learning in which the syllable weight-related constraints do not have sufficient weight to pull stress away from an edge.

8. Discussion: results and future directions

8.1. Results

In this paper, we have pursued what seems to be a novel approach to the study of learning in generative grammar. Prior work takes as given a theory of grammar, and studies its learning (see [Dresher and Kaye, 1990](#) and [Gibson and Wexler, 1994](#) for pioneering work in this tradition). Here we take a given learning theory, and study the consequences of altering the grammatical theory. We have introduced this new way of studying learning and grammar with what seem to us a promising set of initial results. We previewed these results in the introduction of this paper, and now return to them after having presented them in detail.

This paper introduced MaxEnt learning of hidden foot structure by providing learning simulations both for a well-studied test set ([Tesar and Smolensky, 2000, TS](#)), and also for a new test set based on the descriptions of attested languages in StressTyp2 ([Goedemans et al., 2015](#)). We showed that a MaxEnt learner achieved results comparable to the state-of-the-art on the TS test set. We also showed that a set of languages for which learning was particularly difficult fell into a class of iambic nonfinality patterns, which are unattested and whose difficulty is explicable in terms of the structure of the constraint set.

We then turned to the representation and learning of a set of attested stress patterns from StressTyp2. We showed that the original TS constraint set of [Tesar and Smolensky \(2000, TS\)](#), which we refer to as TS-OG, captures a subset of the patterns that a slightly

revised foot-based constraint set does (TS-REV), and also a subset of the patterns that can be generated with a set of grid-based constraints based on [Gordon \(2002\)](#).

In terms of learning, we identified a set of quantity insensitive patterns for which the TS-OG constraints are at an advantage over TS-REV, focusing particularly on the example of Garawa. This serves as a clear example of how explicitly studying learnability with competing grammatical theories can reveal differences that would be hidden if we only compared them in terms of their grammatical coverage. We also showed that adding one of the constraints from TS-OG to the TS-REV constraint set improved learning of the Garawa pattern. This example, along with the comparison of two foot-based constraint sets for Latin, shows that increasing the representational space can aid learning. Larger representational spaces are not always conducive to better learning, of course, and we did find that TS-REV with the added constraint did wind up in local optima in learning Latin more often than the original TS-REV.

Our comparison of learning between foot-based and grid-based constraint sets did not yield a clear winner in overall speed of learning, measured in terms of epochs to convergence. In comparing two grid-based constraint sets, we found that their relative speed differed depending on whether the patterns were quantity sensitive or not, as was the case for the foot-based constraints.

8.2. Future work

All of the constraint sets that we studied here fall short of capturing the full typology that we investigated, and would also fail to represent other patterns that we did not include, such as ones involving ternary distinctions in weight. The development of alternative constraint sets for stress, and the computational assessment of their representational capacity and learnability, is one direction for future work. In addition, the same approach to learning and typology we have taken here could be extended to other linguistic domains, including ones outside of phonology, like syntax (see e.g. [Prickett et al., 2019](#)) on MaxEnt learning of syntax). The software that we have used for this work is freely available, which we hope will aid in pursuing this research.

The learning problems studied here are highly idealized. Dropping those idealizations would require addressing other instances of hidden structure, including syllabification, weight contrasts, and lexical structure allowing for morpheme-specific behavior.

We have assumed a fixed constraint set. Expanding the scope of the learning problems may well be aided by adopting a learner that creates its own constraints, as in [Hayes and Wilson \(2008\)](#), or even a neural network alternative, as in [Prickett and Pater \(2022\)](#). In pursuing neural approaches, their structure might be better understood by comparing their biases to those of learners like those studied here (e.g. do they show evidence of grid- or foot-like representations?).

We have emphasized here that learnability is not always enhanced by limiting the representational space provided by the grammar. But typological restrictiveness can be a goal on its own, and we have at all not considered the extent to which these grammatical theories can represent patterns outside of the attested typology. We expect that they do generate extremely large sets of possible languages, given how

constraint weighting interacts with Alignment constraints (Potts et al., 2010). We see two responses to this. The first would be to abandon typological restrictiveness as a goal, focusing just on learnability, which would be in line with the original learning motivation for Universal Grammar (Chomsky, 1980). The other, presumably more productive response, would be to examine the extent to which unattested languages are less learnable than attested ones, extending the findings on iambic nonfinality in section 3 (see also the prior work cited there on learning biases and stress typology).

Another clear direction for future research is the development of more precise characterizations of the ways in which the structure of the target patterns with each constraint set affects learnability. We have provided explanations for the difficulty of learning several patterns with particular constraint sets, and have noted that the depth of constraint ordering appears to affect learnability. More general and precise formulations of what leads to differences in learnability would help in the quest to enhance learnability in the construction of grammars, and to align learning biases with typological biases.

9. Appendix: StressTyp2 patterns

This is a list of 61 StressTyp2 patterns studied in this paper. For a description of the pattern and source references, replace the last 3 digits in this URL with the three digit code given for the pattern (this URL goes to Latin): http://st2.ullet.net/?content=browse_pattern&pattern=178. The Finite State Transducers can be found in the downloadable archive at <http://st2.ullet.net/?content=page&id=19>. Files needed to replicate the learning and Linear Programming analyses for each pattern can be found in Lee and Pater (2024); these also use the StressTyp2 three-digit codes provided here.

1. final (111) QI
2. initial (112) QI
3. second (113) QI
4. Aklan (114) QS
5. penultimate (115) QI
6. leftmost heavy (116) QS
7. penult alt secondary (117) QI
8. initial alt secondary (118) QI
9. penult initial secondary (119) QI
10. Aranda Western (127) QS
11. second alt secondary (128) QI
12. final initial secondary (129) QI
13. final alt secondary (131) QI
14. penult if heavy else final if heavy else penult (132) QS
15. initial alt secondary left (133) QI
16. Bhojpuri SPD (134) QI
17. Biangai (136) QI
18. Buriat (139) QS
19. Cayuvava (144) QI
20. Central Alaskan Yupik (145) QI

21. Cheremis Eastern (146) QS
22. rightmost nonfinal heavy else final (148) QS
23. rightmost heavy else initial (149) QS
24. antepenult (150) QI
25. rightmost heavy else penult (153) QS
26. final if heavy weighted secondary (156) QS
27. Garawa (157) QI
28. Georgian SPD (158) QI
29. rightmost heavy else final (159) QS
30. first if heavy else second if heavy else first weighted secondary (162) QS
31. rightmost heavy before final else penult (165) QS
32. first if heavy else second (167) QS
33. Icuá Tupi (169) QI
34. Indonesian (170) QI
35. Ioway Oto (171) QI
36. penult if heavy else final (172) QS
37. leftmost heavy else final (176) QS
38. Kuuku-Ya?u SPD (177) QS
39. Latin Classical (178) QS
40. penult alt secondary no antepenult (179) QI
41. Maithili (182) QS
42. Malay (184) QS
43. Mingrelian (191) QI
44. Miwok Sierra (192) QS
45. Mongolian Khalkha SPD (194) QS
46. Mongolian Khalkha (196) QS
47. Nubian Dongolese (199) QS
48. Nywaygi (200) QS
49. Romanian nouns SPD (203) QS
50. Romansh Berguener SPD (204) QS
51. Seminole Creek (205) QS
52. Shoshone Gosiute (208) QI
53. Shoshone Tumpisa (209) QS
54. Sindhi SPD (210) QS
55. Sorbian (211) QI
56. Southern Paiute (212) QI
57. Tauya (213) QI
58. Tongan SPD (214) QS
59. Udihe (215) QI
60. Yapese SPD (218) QS
61. Yidin SPD (219) QS

Acknowledgments. Thank you to Alessa Farinella and Cerys Hughes for their collaboration on an earlier presentation of some of this material, and to Bruce Hayes, Gaja Jarosz, Andrew Lamont and Aleksei Nazarov for discussion. Thanks to Paul Boersma for sharing the Tesar and Smolensky (2000) test set, as well as Bruce Tesar for making the test set available. We also thank audiences at the 2019 Manchester Phonology Meeting

and the Annual Meetings on Phonology in 2021 and 2022 for their feedback, as well as the UMass Sound Workshop.

Funding statement. This research was supported by NSF grant BCS-2140826 to the University of Massachusetts Amherst.

Supplementary material. An Excel file containing supplementary material discussed in the text can be downloaded at [this URL](#).

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